

SonarQube Analysis for Angular Calculator Application

Objective

The objective of this project was to perform code quality analysis for an Angular Calculator application using SonarQube. A separate Build Server was used to install Node.js and npm, while another server was configured as the SonarQube Server to host the SonarQube dashboard and analysis reports.

Architecture Used

1. Build Server

- Install Node.js and npm
- Clone Angular Calculator source code
- Create SonarQube configuration file
- Run SonarQube analysis

2. SonarQube Server

- Install Java
- Install and start SonarQube
- View project analysis reports on port 9000

Step 1: Install Node.js on Build Server

Run the following command:

```
sudo apt install nodejs
```

```
ubuntu@Build-VM:~$ sudo apt install nodejs
Installing:
  nodejs

Installing dependencies:
  libada-url0-3  libnode127  node-acorn  node-cjs-module-lexer  node-llhttp
  libcares2      libsimdjson29  node-balanced-match  node-corepack  node-lru-cache
  libllhttp9.3  libsimdutf31  node-brace-expansion  node-debug  node-minima

Suggested packages:
  npm

Summary:
  Upgrading: 0, Installing: 21, Removing: 0, Not Upgrading: 6
  Download size: 25.8 MB
  Space required: 106 MB / 2048 MB
```

Step 2: Install npm on Build Server

Run the following command:

```
sudo apt install npm
```

```
ubuntu@Build-VM:~$ sudo apt install npm
Installing:
  npm

Installing dependencies:
  alacritty          node-gyp
  build-essential    node-has-flag
  bzip2              node-has-unicode
  cpp                node-has-value
  cpp-15             node-has-values
  cpp-15-x86-64-linux-gnu node-hosted-git-info
  cpp-x86-64-linux-gnu node-http-proxy-agent
  dpkg-dev           node-https-proxy-agent
  eslint             node-iconv-lite
  fakeroot            node-ipc-utile
```

Step 3: Clone Angular Calculator Project

Clone the project using:

```
git clone https://github.com/Akiraared/AngularCalculator.git
```

Move into the project directory:

```
cd AngularCalculator
```

```
ubuntu@Build-VM:~$ git clone https://github.com/Akiranred/AngularCalculator.git
Cloning into 'AngularCalculator'...
remote: Enumerating objects: 39, done.
remote: Total 39 (delta 0), reused 0 (delta 0), pack-reused 39 (from 1)
Receiving objects: 100% (39/39), 107.66 KiB | 8.28 MiB/s, done.
ubuntu@Build-VM:~$ ls
AngularCalculator  Java-Web-Calculator-App
ubuntu@Build-VM:~$ cd AngularCalculator/
```

Step 4: Install Angular CLI

Install Angular CLI globally:

```
sudo npm install -g @angular/cli
```

```
ubuntu@Build-VM:~/AngularCalculator$ sudo npm install -g @angular/cli
added 288 packages in 9s

70 packages are looking for funding
  run `npm fund` for details
ubuntu@Build-VM:~/AngularCalculator$ npm
npm <command>

Usage:

npm install          install all the dependencies in your project
npm install <foo>    add the <foo> dependency to your project
npm test             run this project's tests
npm run <foo>        run the script named <foo>
npm <command> -h     quick help on <command>
```

Step 5: Setup SonarQube Server

Check Java version:

```
java --version
```

```
ubuntu@Sonar-VM:~$ java --version
openjdk 17.0.18 2026-01-20
OpenJDK Runtime Environment (build 17.0.18+8-Ubuntu-1)
OpenJDK 64-Bit Server VM (build 17.0.18+8-Ubuntu-1, mixed mode, sharing)
```

Download SonarQube:

wget <https://binaries.sonarsource.com/Distribution/sonarqube/sonarqube-10.6.0.92116.zip>

```
ubuntu@Sonar-VM:~$ sudo wget https://binaries.sonarsource.com/Distribution/sonarqube/sonarqube-10.6.0.92116.zip
--2026-05-20 04:54:18-- https://binaries.sonarsource.com/Distribution/sonarqube/sonarqube-10.6.0.92116.zip
Resolving binaries.sonarsource.com (binaries.sonarsource.com)... 3.167.37.95, 3.167.37.88, 3.167.37.10, ...
Connecting to binaries.sonarsource.com (binaries.sonarsource.com)|3.167.37.95|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 731034761 (697M) [binary/octet-stream]
Saving to: 'sonarqube-10.6.0.92116.zip'

sonarqube-10.6.0.92116.zip  100%[=====] 697.17M  435MB/s  in 1.6s

2026-05-20 04:54:19 (435 MB/s) - 'sonarqube-10.6.0.92116.zip' saved [731034761/731034761]
```

Step 6: Start SonarQube

Extract and start SonarQube:

unzip sonarqube-10.6.0.92116.zip

```
ubuntu@Sonar-VM:~$ sudo unzip sonarqube-10.6.0.92116.zip
Archive:  sonarqube-10.6.0.92116.zip
  creating: sonarqube-10.6.0.92116/
  creating: sonarqube-10.6.0.92116/jres/
  inflating: sonarqube-10.6.0.92116/jres/OpenJDK17U-jre_x64_linux_hotspot_17.0.11_9.tar.gz
  inflating: sonarqube-10.6.0.92116/jres/OpenJDK17U-jre_aarch64_linux_hotspot_17.0.11_9.tar.gz
  inflating: sonarqube-10.6.0.92116/jres/OpenJDK17U-jre_x64_alpine-linux_hotspot_17.0.11_9.tar.gz
```

cd sonarqube/bin/linux-x86-64

./sonar.sh start

Check status:

./sonar.sh status

```
ubuntu@Sonar-VM:~$ cd sonarqube/
ubuntu@Sonar-VM:~/sonarqube$ ls
COPYING  bin  conf  data  dependency-license.json  elasticsearch  extensions  jres  lib
ubuntu@Sonar-VM:~/sonarqube$ cd bin/
ubuntu@Sonar-VM:~/sonarqube/bin$ ls
elasticsearch  linux-x86-64  macosx-universal-64  windows-x86-64  winsw-license
ubuntu@Sonar-VM:~/sonarqube/bin$ cd linux-x86-64/
ubuntu@Sonar-VM:~/sonarqube/bin/linux-x86-64$ ls
SonarQube.pid  sonar.sh
ubuntu@Sonar-VM:~/sonarqube/bin/linux-x86-64$ ./sonar.sh status
/usr/bin/java
Removed stale pid file: ./SonarQube.pid
SonarQube is not running.
ubuntu@Sonar-VM:~/sonarqube/bin/linux-x86-64$ ./sonar.sh start
/usr/bin/java
Starting SonarQube...
Started SonarQube.
ubuntu@Sonar-VM:~/sonarqube/bin/linux-x86-64$ ./sonar.sh status
/usr/bin/java
SonarQube is running (2526).
```

Step 7: Create SonarQube Token

Login to SonarQube dashboard and generate a token from:

Administration → Security → Users → Tokens

Token - `squ_06abeefd6bfaf47b4b169a227ab50891c3e39e4e`

Tokens of *Administrator*

Name	Type	Project	Last use	Created	Expiration	
sonar	User		< 1 hour ago	May 20, 2026	June 19, 2026	Revoke

Step 8: Create sonar-property.js File

Create a file named `sonar-property.js` inside the `AngularCalculator` project directory and add the configuration for SonarQube connection.

```
ubuntu@Build-VM:~/AngularCalculator$ cat sonar-property.js
const sonarqubeScanner = require('sonarqube-scanner');

sonarqubeScanner.default(
{
  serverUrl: 'http://54.90.67.193:9000',

  token: 'squ_06abeefd6bfaf47b4b169a227ab50891c3e39e4e',

  options:
  {
    'sonar.projectKey': 'AngularCalculator',
    'sonar.projectName': 'AngularCalculator',
    'sonar.projectVersion': '1.0',
    'sonar.sources': 'src',
    'sonar.sourceEncoding': 'UTF-8'
  }
},
() => {
  console.log('Scan completed successfully');
}
);
```

Step 9: Run SonarQube Analysis

Run the following command from the AngularCalculator directory:

```
node sonar-property.js
```

```
ubuntu@Build-VM:~/AngularCalculator$ node sonar-property.js
[INFO] Bootstrapper: Retrieving info from "package.json" file
[INFO] Bootstrapper: Platform: linux x64
[INFO] Bootstrapper: Server URL: http://54.90.67.193:9000
[INFO] Bootstrapper: Version: 4.3.6
[INFO] Bootstrapper: SonarQube server version: 10.6.0
[INFO] Bootstrapper: JRE provisioning is supported
[INFO] Bootstrapper: Using JRE from the cache
[INFO] ScannerEngine: Starting SonarScanner Engine...
[INFO] ScannerEngine: Java 17.0.11 Eclipse Adoptium (64-bit)
[INFO] ScannerEngine: Load global settings
[INFO] ScannerEngine: Load global settings (done) | time=170ms
[INFO] ScannerEngine: SonarQube: 10.6.0
[INFO] ScannerEngine: ANALYSIS SUCCESSFUL, you can find the results at: http://54.90.67.193:9000/dashboard
[INFO] ScannerEngine: Note that you will be able to access the updated dashboard once the server has processed analysis report
[INFO] ScannerEngine: More about the report processing at http://54.90.67.193:9000/api/ce/task?id=e71e39b3
```

If successful, the analysis will complete and the results will appear in the SonarQube dashboard.

Step 10: View Results

Open the SonarQube dashboard in browser:

```
http://54.90.67.193:9000
```

The screenshot shows the SonarQube web interface in a browser. The address bar indicates the URL is `http://54.90.67.193:9000/dashboard?id=AngularCalculator&codeScope=overall`. A warning banner at the top states: "You're running a version of SonarQube that is no longer active. Please upgrade to an active version immediately." with a "Learn More" button. The navigation bar includes links for Projects, Issues, Rules, Quality Profiles, Quality Gates, Administration, and More. The main content area shows the project "AngularCalculator" with a dropdown menu set to "main" (indicated by a green checkmark). Below this, the "Overview" tab is selected, showing a "Quality Gate" status of "Passed" with a green checkmark icon. The text "286 Lines of Code · Version 1.0" is visible. At the bottom, there are tabs for "New Code" and "Overall Code".

The dashboard will display:

- Code Quality
- Bugs
- Vulnerabilities
- Code Smells
- Quality Gate Status

Conclusion:

Successfully configured a multi-server SonarQube environment for analyzing an Angular Calculator application. Node.js, npm, and Angular dependencies were installed on the Build Server, while SonarQube was installed on a dedicated SonarQube Server. Using the sonar-property.js configuration file, the Angular project was analyzed successfully and the code quality report was viewed in the SonarQube dashboard.